FINTECH CUSTOMER RISK & BEHAVIOUR INTELLIGENCE

Customer Segmentation, Churn Prediction & Credit Behaviour Analytics

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EXECUTIVE SUMMARY

Customer retention is one of the most critical challenges in the banking and financial services industry. Customer attrition not only reduces revenue but also increases customer acquisition costs. This project aims to analyze customer behavior, identify churn patterns, segment customers based on their financial activities, predict churn risk using machine learning, estimate customer credit behavior, and generate actionable business recommendations.

The analysis was conducted using the Bank Churners dataset containing information on 10,127 customers and their demographic, transactional, and credit-related characteristics.

Key Results

- Total Customers: 10,127
- Existing Customers: 8,500 (83.93%)
- Attrited Customers: 1,627 (16.07%)
- Optimal Customer Segments: 4
- Best Churn Prediction Model: XGBoost
- Best ROC-AUC Score: 0.9921
- Most Important Churn Driver: Total Transaction Amount
- Income Category significantly impacts churn ($p = 0.0250$)
- Transaction behavior significantly differs between churned and retained customers ($p = 2.719 \times 10^{-112}$)

1. INTRODUCTION

Customer churn is a major concern in the banking industry because retaining existing customers is significantly cheaper than acquiring new ones. Understanding customer behavior and predicting churn risk enables financial institutions to proactively engage customers and improve retention rates.

The objective of this project is to:

- Understand customer behavior patterns
- Identify customer segments
- Predict churn risk
- Predict customer credit behavior
- Perform statistical analysis on customer characteristics
- Generate business recommendations

2. DATASET OVERVIEW

The dataset contains information about credit card customers including demographic attributes, transaction behavior, account activity, and churn status.

Dataset Statistics

Metric	Value
Total Records	10,127
Total Features	23
Existing Customers	8,500
Attrited Customers	1,627

	CLIENTNUM	Attrition_Flag	Customer_Age	Gender	Dependent_count	Education_Level	Marital_Status	Income_Category	Card_Category	Months_on_book
0	768805383	Existing Customer	45	M	3	High School	Married	60K−80K	Blue	39
1	818770008	Existing Customer	49	F	5	Graduate	Single	Less than $40K	Blue	44
2	713982108	Existing Customer	51	M	3	Graduate	Married	80K−120K	Blue	36
3	769911858	Existing Customer	40	F	4	High School	Unknown	Less than $40K	Blue	34
4	709106358	Existing Customer	40	M	3	Uneducated	Married	60K−80K	Blue	21

5 rows × 23 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10127 entries, 0 to 10126
Data columns (total 23 columns):
 #   Column                                                                                                                                Non
-Null Count  Dtype
---  ------                                                                                                                                ---
-----------  -----
 0   CLIENTNUM                                                                                                                             101
27 non-null  int64
 1   Attrition_Flag                                                                                                                        101
27 non-null  object
 2   Customer_Age                                                                                                                          101
27 non-null  int64
 3   Gender                                                                                                                                101
27 non-null  object
 4   Dependent_count                                                                                                                       101
27 non-null  int64
 5   Education_Level                                                                                                                       101
27 non-null  object
 6   Marital_Status                                                                                                                        101
27 non-null  object
 7   Income_Category                                                                                                                       101
27 non-null  object
 8   Card_Category                                                                                                                         101
27 non-null  object
 9   Months_on_book                                                                                                                        101
27 non-null  int64
 10  Total_Relationship_Count                                                                                                              101
27 non-null  int64
```

3. EXPLORATORY DATA ANALYSIS

3.1 Missing Value Analysis

The dataset was examined for missing values before conducting further analysis.

Findings

No missing values were detected in the dataset, indicating excellent data quality.

```
df.isnull().sum()
```

```
Attrition_Flag              0
Customer_Age                0
Gender                      0
Dependent_count             0
Education_Level             0
Marital_Status              0
Income_Category             0
Card_Category               0
Months_on_book              0
Total_Relationship_Count    0
Months_Inactive_12_mon      0
Contacts_Count_12_mon       0
Credit_Limit                0
Total_Revolving_Bal         0
Avg_Open_To_Buy             0
Total_Amt_Chng_Q4_Q1        0
Total_Trans_Amt             0
Total_Trans_Ct              0
Total_Ct_Chng_Q4_Q1         0
Avg_Utilization_Ratio       0
dtype: int64
```

3.2 Churn Distribution Analysis

Understanding churn distribution is essential because it reveals class imbalance and helps determine appropriate evaluation metrics.

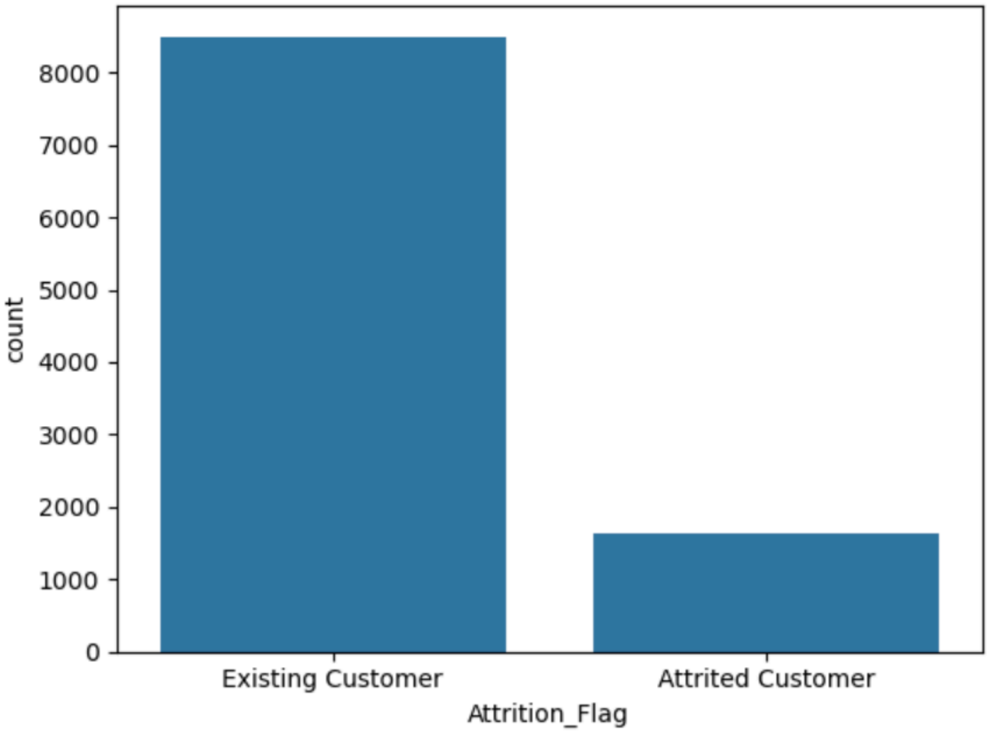
Findings

- Existing Customers: 83.93%
- Attrited Customers: 16.07%

This indicates a moderate class imbalance where retained customers significantly outnumber churned customers.

```
#Churn Distribution

sns.countplot(x='Attrition_Flag', data=df)
plt.show()
```

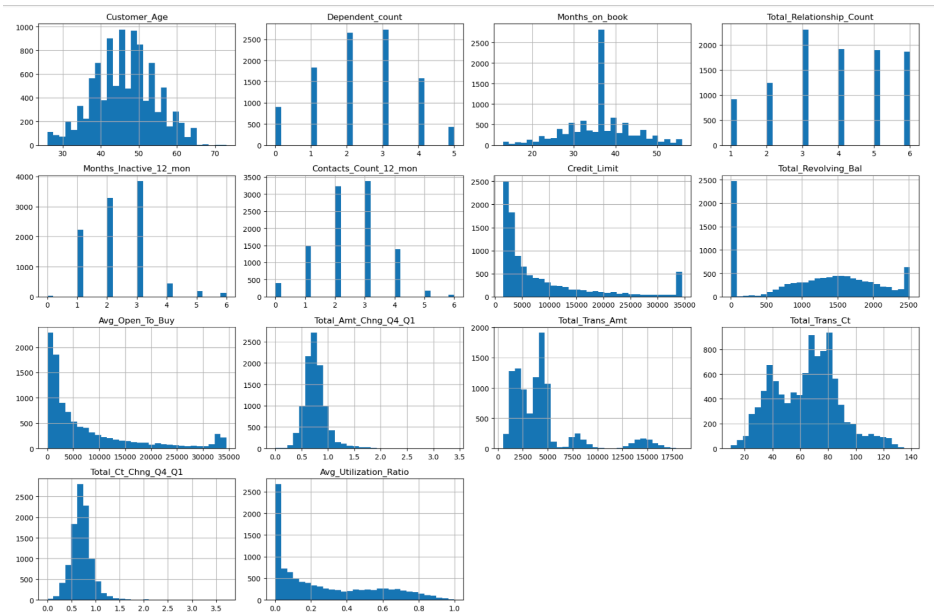


Business Interpretation

Most customers remain active; however, the 16% churn rate represents a significant revenue risk for the bank.

3.3 Feature Distribution Analysis

Histograms were generated for all numerical variables to understand their distributions and identify skewness patterns.

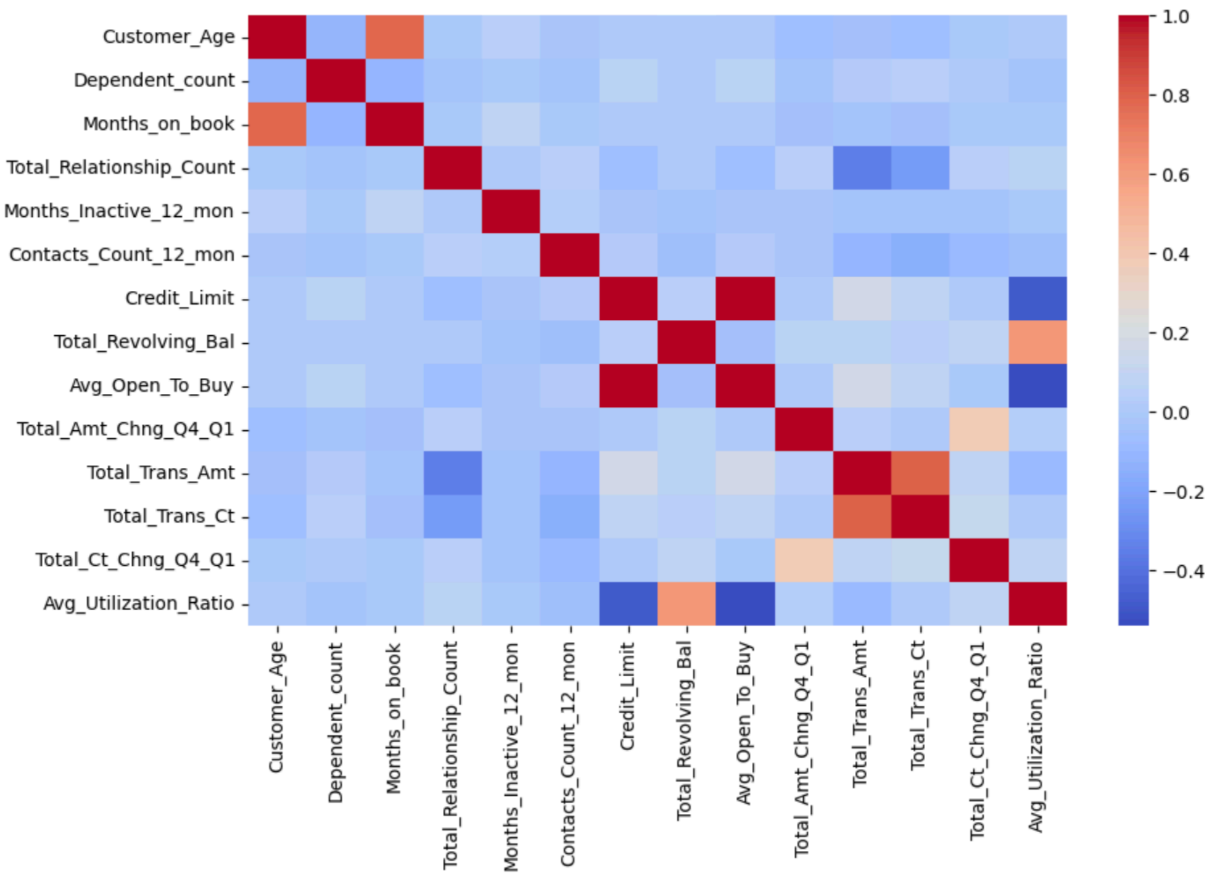


Findings

Several financial variables such as Credit Limit and Transaction Amount exhibited right-skewed distributions, indicating the presence of high-value customers.

3.4 Correlation Analysis

A correlation heatmap was created to identify relationships between numerical variables.



Findings

Strong relationships were observed among:

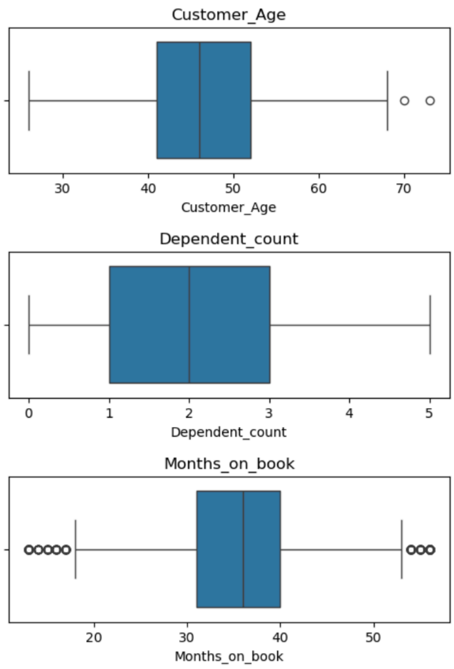
- Credit Limit and Transaction Amount
- Transaction Count and Transaction Amount
- Revolving Balance and Utilization Ratio

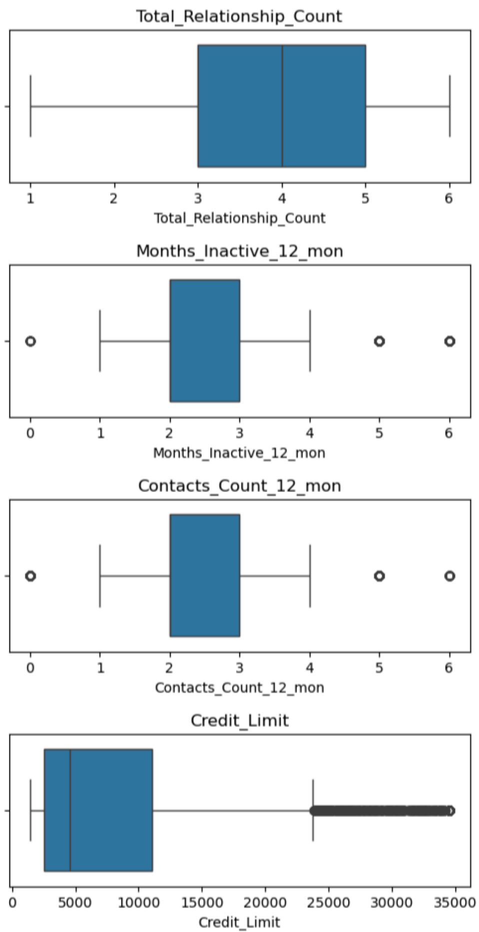
Business Interpretation

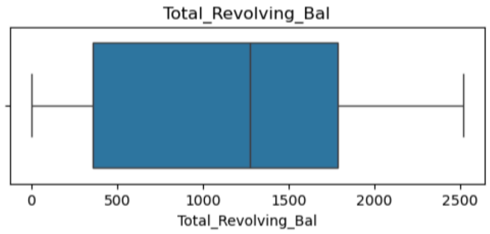
Customers with higher transaction activity tend to possess higher credit limits and generate more revenue for the bank.

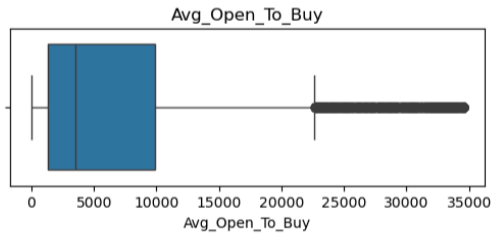
3.5 Outlier Analysis

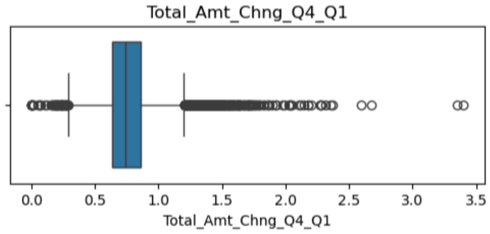
Boxplots were used to detect potential outliers.

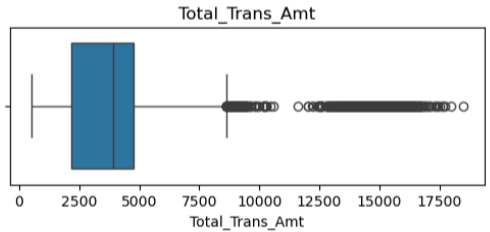


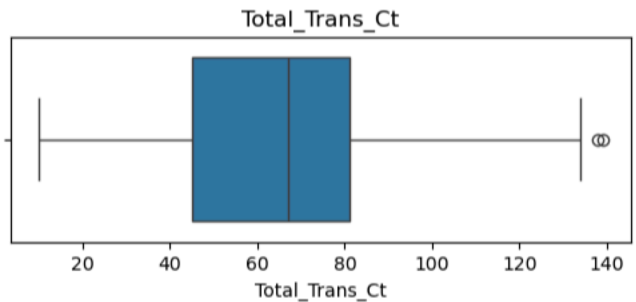


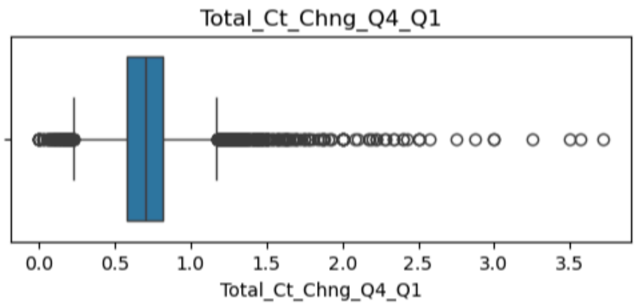

Total_Revolving_Bal
0
500
1000
1500
2000
2500
Total_Revolving_Bal

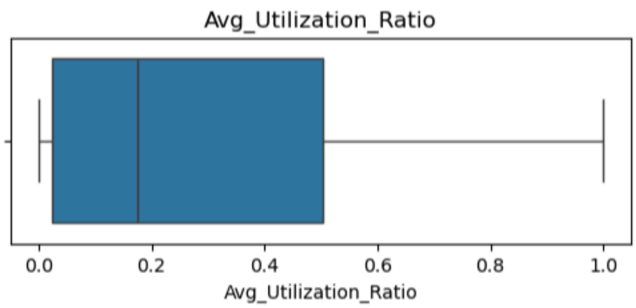

Avg_Open_To_Buy
0
5000
10000
15000
20000
25000
30000
35000
Avg_Open_To_Buy


Total_Amt_Chng_Q4_Q1
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Total_Amt_Chng_Q4_Q1


Total_Trans_Amt
0
2500
5000
7500
10000
12500
15000
17500
Total_Trans_Amt


Total_Trans_Ct
20
40
60
80
100
120
140
Total_Trans_Ct


Total_Ct_Chng_Q4_Q1
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
Total_Ct_Chng_Q4_Q1


Avg_Utilization_Ratio
0.0
0.2
0.4
0.6
0.8
1.0
Avg_Utilization_Ratio

Findings

Several variables contained extreme values. These observations were retained because they represent genuine high-value banking customers rather than data errors.

4. DATA PREPROCESSING

Categorical variables were transformed into numerical format using Label Encoding. Feature scaling was performed using StandardScaler to ensure all variables contributed equally during clustering.

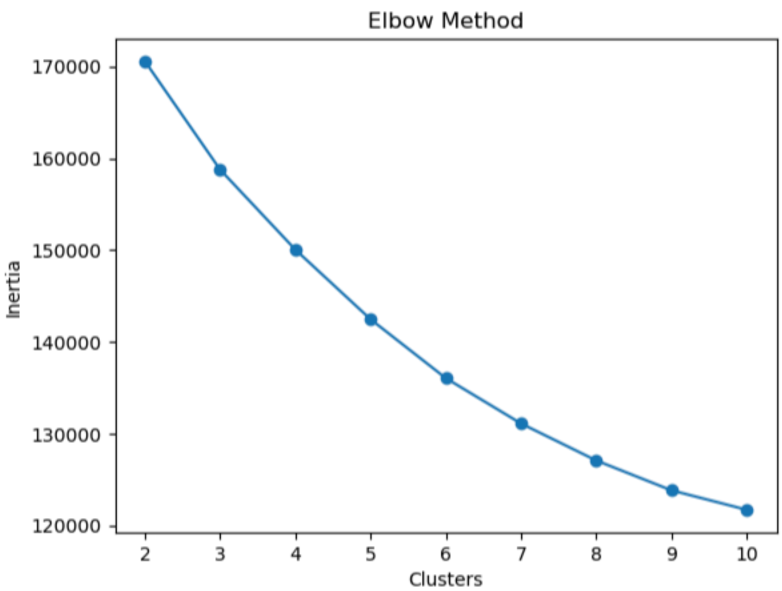
5. CUSTOMER SEGMENTATION USING K-MEANS

Objective

The objective of customer segmentation is to group customers with similar financial behavior patterns.

5.1 Elbow Method

The Elbow Method was used to determine the optimal number of clusters.



5.2 Silhouette Analysis

Silhouette Scores were calculated to evaluate clustering quality.

K	Silhouette Score
2	0.1744
3	0.0870
4	0.0958
5	0.0989

Although K=2 achieved the highest score, K=4 was selected because it produced more meaningful business segments.

5.3 Customer Cluster Profiling

Cluster 0 – Loyal Everyday Customers

Characteristics:

- Credit Limit: 3,574
- Transaction Amount: 3,562
- Transaction Count: 63

Business Meaning:

These customers actively use their cards and exhibit stable spending patterns.

Cluster 1 – Low Engagement Customers

Characteristics:

- Credit Limit: 6,773
- Transaction Amount: 3,164
- Utilization Ratio: 0.078

Business Meaning:

Customers have access to credit but demonstrate low card utilization, indicating potential churn risk.

Cluster 2 – Premium Credit Customers

Characteristics:

- Credit Limit: 27,513
- Transaction Amount: 4,592
- Transaction Count: 65

Business Meaning:

High-credit customers with strong financial profiles and premium banking potential.

Cluster 3 – High Value Power Users

Characteristics:

- Credit Limit: 10,944
- Transaction Amount: 12,619
- Transaction Count: 104

Business Meaning:

Most profitable customer group generating substantial transaction revenue.

Cluster	Customer_Age	Dependent_count	Months_on_book	Total_Relationship_Count	Months_Inactive_12_mon	Contacts_Count_12_mon	Credit_Limit	Total_
0	46.295229	2.312376	35.731362	4.079771	2.309394	2.345676	3574.162276	
1	46.575853	2.317323	36.312073	3.977690	2.423097	2.634121	6772.588215	
2	46.257485	2.546407	35.827844	3.750749	2.287425	2.478293	27512.522455	
3	45.555904	2.323929	35.369906	2.118077	2.223615	2.172414	10943.652038	

6. CHURN PREDICTION

Three machine learning models were developed to predict customer churn.

6.1 Logistic Regression

ROC-AUC Score: 0.9079

Business Interpretation

Logistic Regression provided a strong baseline model and demonstrated that churn can be predicted effectively using customer behavior data.

6.2 Random Forest

ROC-AUC Score: 0.9880

Business Interpretation

Random Forest successfully captured complex customer behavior patterns and significantly improved predictive performance.

6.3 XGBoost

ROC-AUC Score: 0.9921

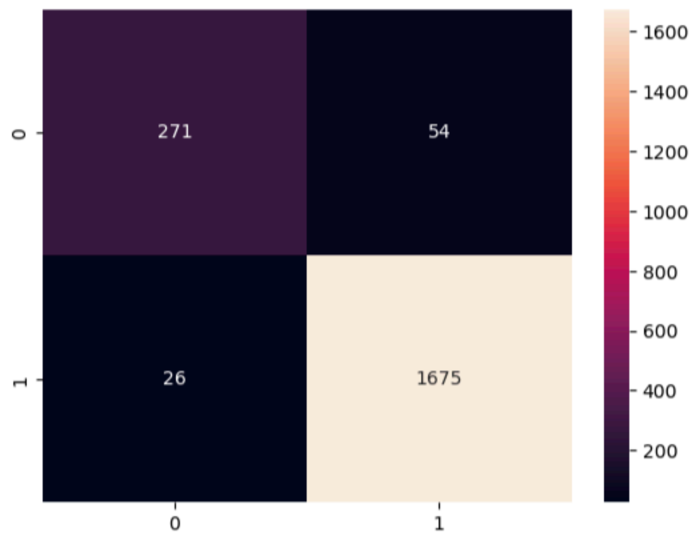
Business Interpretation

XGBoost achieved the highest predictive accuracy and emerged as the best-performing model.

Model Comparison

Model	ROC-AUC
Logistic Regression	0.9079
Random Forest	0.9880
XGBoost	0.9921

Confusion Matrix



7. FEATURE IMPORTANCE ANALYSIS

The Random Forest model was used to identify the strongest drivers of customer churn.

Rank	Feature	Importance
1	Total_Trans_Amt	0.1855
2	Total_Trans_Ct	0.1759
3	Total_Ct_Chng_Q4_Q1	0.1147
4	Total_Revolving_Bal	0.1046
5	Total_Relationship_Count	0.0673

	Feature	Importance
15	Total_Trans_Amt	0.185540
16	Total_Trans_Ct	0.175931
17	Total_Ct_Chng_Q4_Q1	0.114711
12	Total_Revolving_Bal	0.104601
8	Total_Relationship_Count	0.067335
18	Avg_Utilization_Ratio	0.064555
14	Total_Amt_Chng_Q4_Q1	0.061048
0	Customer_Age	0.033522
13	Avg_Open_To_Buy	0.031371
11	Credit_Limit	0.030922
10	Contacts_Count_12_mon	0.027803
7	Months_on_book	0.024661
9	Months_Inactive_12_mon	0.023772
2	Dependent_count	0.012357
3	Education_Level	0.011628

Business Interpretation

Transaction-related variables are the strongest indicators of churn risk.

Customers showing declining transaction activity require immediate intervention.

8. CREDIT BEHAVIOUR PREDICTION

Linear Regression, Ridge Regression, and Lasso Regression models were developed to estimate customer credit limits.

Business Interpretation

Customer financial behavior can be used to estimate credit capacity and support credit risk management decisions.

9. HYPOTHESIS TESTING

Test 1: Income Category vs Churn

Statistical Test: Chi-Square Test

Hypotheses

H0: Churn is independent of Income Category.

H1: Churn depends on Income Category.

Results

p-value = 0.0250

Decision

Since $p < 0.05$, H0 is rejected.

Conclusion

Income Category significantly influences customer churn.

Test 2: Transaction Amount vs Churn

Statistical Test: Mann-Whitney U Test

Hypotheses

H0: Transaction distributions are identical.

H1: Transaction distributions differ.

Results

p-value = 2.719×10^{-112}

Decision

Since $p < 0.05$, H0 is rejected.

Conclusion

Transaction behavior differs significantly between churned and retained customers.

10. BUSINESS RECOMMENDATIONS

Recommendation 1: Churn Prevention Program

Target Segment:

Cluster 1 (Low Engagement Customers)

Actions:

- Cashback offers
- Personalized discounts
- Retention campaigns

Recommendation 2: Premium Upselling

Target Segment:

Cluster 2 (Premium Credit Customers)

Actions:

- Premium credit cards
- Wealth management products
- Higher credit limits

Recommendation 3: Protect High Value Customers

Target Segment:

Cluster 3 (Power Users)

Actions:

- Loyalty programs
- Exclusive benefits
- Dedicated customer support

Recommendation 4: Early Warning Churn System

Monitor:

- Transaction Amount
- Transaction Count
- Relationship Count

These variables are the strongest predictors of churn.

CONCLUSION

This project successfully combined exploratory data analysis, customer segmentation, churn prediction, regression modeling, and statistical testing to generate actionable insights for the banking sector.

The analysis revealed that transaction behavior is the most influential factor affecting customer churn. Four distinct customer segments were identified, enabling targeted business strategies. Among the classification models tested, XGBoost achieved the highest predictive performance with a ROC-AUC score of 0.9921.

The findings of this project can assist financial institutions in improving customer retention, enhancing profitability, optimizing marketing strategies, and supporting data-driven decision-making.